

AI-POWERED VIRTUAL ASSISTANT FOR THE MSBA WEBSITE

Presenter : Sneha More

Academic Supervisor : Prof. Pang Yan James

PROJECT BACKGROUND

Context

NUS MSBA programme receives a large number of inquiries from prospective students globally.

Current Process

Queries are handled manually via email and static FAQ pages.

Challenges

Time-consuming, inconsistent responses, and limited scalability.

Need

A streamlined, accurate, and scalable solution for delivering programme information in real-time.

KEY DELIVERABLE

A RAG-powered AI chatbot built on Google Vertex AI Agent Builder that delivers accurate, grounded, and real-time responses to MSBA programme queries, reducing manual workload while ensuring consistency and scalability.



High
Volume of
Queries



Manual
Handling via
Email/FAQ



RAG-powered chatbot
(Google Vertex AI Agent Builder)



Accurate, Fast &
Grounded Responses

KEY COMPONENTS

Data Ingestion & Parsing

Automated ingestion of brochure, FAQ, and scraped website content with structured parsing and cleanup to ensure consistent formats for downstream processing.

Chunking & Embedding Engine

Text split into ~500-token chunks with semantic overlap for contextual integrity. Embeddings generated using Vertex AI for high-precision semantic retrieval.

Retrieval-Augmented Generation (RAG)

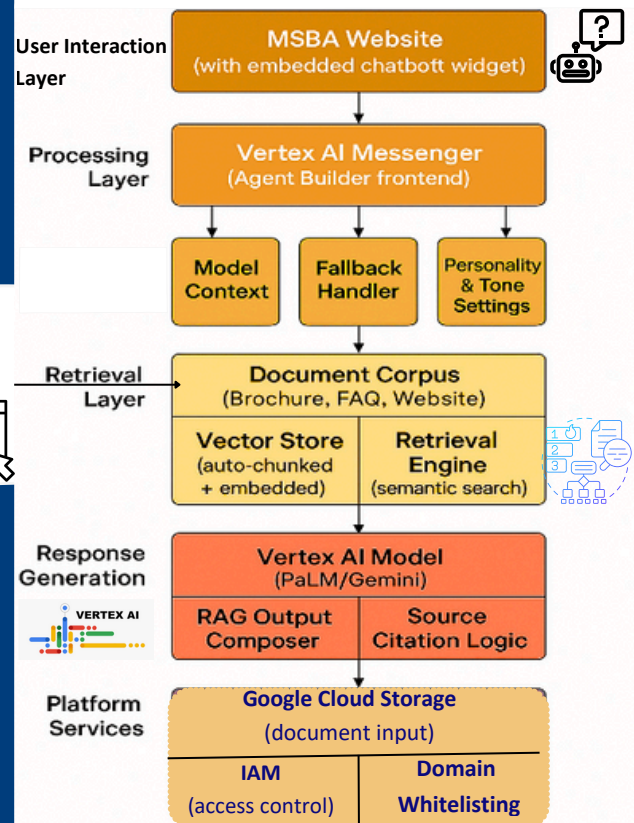
Top-K retrieval with grounding to the institutional knowledge base, ensuring responses remain accurate, domain-restricted, and fully cited.

Fallback & Conversation Management

Generative fallback pipeline for unmatched queries, **conversation logging** for analytics, and integrated user experience **feedback loop** for continuous improvement.



TECHNICAL ARCHITECTURE



IMPACT

- Improved accessibility and reliability of program information for prospective students
- Enabled faster query resolution while reducing manual intervention by administrative staff.
- Reduced misinformation risk, increased user trust through domain-grounded responses with verifiable citations.
- Provided a scalable framework for multilingual & analytics-driven support for future AI adoption.